

L0 · inputs

5 descriptors:
location · building
population · equip.
noise + seed s

L1 · resolution

69 typed knobs;
CLI > scenario
> descriptor
> default;
provenance per knob

L2 · seeding

SeedSequence(s ,
SHA-256(node));
hash-keyed streams
order-independent

L3 · sampling DAG

Tier 1 · roots

C A S O T U F X

Tier 1.5 · per-entity

$A_{user}(\phi, \kappa, \delta) \cdot A_{fleet}$

Tier 2 · latents

EnergyPlus load · PV
 $f_{arr} f_{dwell} f_{soc}$ + copula

Tier 3 · renderers

10 output tables

L4 · noise

optional,
output-side;
clean = 0 jitter;
repairs logged

L5 · validation

59 invariants
in 9 families;
hard errors
reject the run

weather channel (one realization)

TMYx EPW → perturb: $+\Delta T \cdot \times \text{solar} \cdot$
 $+\Delta \text{dew (RH recomputed)} \cdot \times \text{wind}$
→ one weather frame per sample

*EnergyPlus
+ PV model*

exported as weather_data.csv (same frame)

L6 · outputs

10 CSV tables
+ manifest.json
(checksums,
knob sources,
verdict)